

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**Product Identifier**

**Product Description:** Formic acid, OPTIMA LC/MS Grade (99.5%)  
**Cat No. :** A117-50; A117-10X1AMP; A117-1AMP; A117-05AMP; A117-2AMP; NC1450425; XXA117100ML; NC1484204; XXA1171LI  
**Synonyms** Methanoic acid; FA (OPTIMA LC/MS)  
**CAS No** 64-18-6  
**Molecular Formula** C H2 O2

**Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**Details of the supplier of the safety data sheet**

|                                                                                                                                                                                               |                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Importer</b><br>Fisher Scientific Korea<br>D5,D6, Incheon Airport Logistics Complex<br>150, Gonghangdong-Ro 296 Beon-Gil<br>Jung-Gu, Incheon<br>Tel: +82-1661-9555<br>Fax: +82-2-2023-0603 | <b>Supplier</b><br>Fisher Scientific<br>One Reagent Lane<br>Fair Lawn, NJ 07410<br>Tel: (201) 796-7100 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|

**E-mail address** Chem.KR@thermofisher.com

**Emergency Telephone Number**

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970  
 CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

## SECTION 2: HAZARDS IDENTIFICATION

**Classification of the substance or mixture**
**Physical hazards**

Flammable liquids Category 3

**Health hazards**

|                                    |              |
|------------------------------------|--------------|
| Acute Oral Toxicity                | Category 4   |
| Acute Inhalation Toxicity - Vapors | Category 3   |
| Skin Corrosion/Irritation          | Category 1 A |
| Serious Eye Damage/Eye Irritation  | Category 1   |

**Environmental hazards**

Based on available data, the classification criteria are not met

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## Label Elements



**Signal Word**

**Danger**

### **Hazard Statements**

H226 - Flammable liquid and vapor  
H302 - Harmful if swallowed  
H314 - Causes severe skin burns and eye damage  
H331 - Toxic if inhaled  
H318 - Causes serious eye damage

### **Precautionary Statements**

#### **Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P264 - Wash hands and face thoroughly after handling  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P271 - Use only outdoors or in a well-ventilated area  
P233 - Keep container tightly closed  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P270 - Do not eat, drink or smoke when using this product  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray

#### **Response**

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell  
P330 - Rinse mouth  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P311 - Call a POISON CENTER or doctor  
P321 - Specific treatment (see supplemental first aid instructions on this label)  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction  
P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P363 - Wash contaminated clothing before reuse  
P310 - Immediately call a POISON CENTER or doctor

#### **Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up  
P403 + P235 - Store in a well-ventilated place. Keep cool

#### **Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

Lachrymator (substance which increases the flow of tears)  
This product does not contain any known or suspected endocrine disruptors

## NFPA

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Health  
3

Flammability  
2

Instability  
1

Physical hazards  
N/A

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component   | Common Name    | CAS No  | Index No | Weight % |
|-------------|----------------|---------|----------|----------|
| Formic acid | Methanoic acid | 64-18-6 | KE-17233 | 99 - 100 |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

#### Eye Contact

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. If not breathing, give artificial respiration.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Causes burns by all exposure routes. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated.

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## Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen, Thermal decomposition can lead to release of irritating gases and vapors.

## Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Keep away from heat, sparks and flame. Containers should be vented periodically in order to overcome pressure buildup.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component   | CAS No  | Korea                                  | ACGIH TLV                  | OSHA PEL                                                                                             |
|-------------|---------|----------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------|
| Formic acid | 64-18-6 | TWA: 5 ppm<br>TWA: 9 mg/m <sup>3</sup> | TWA: 5 ppm<br>STEL: 10 ppm | (Vacated) TWA: 5 ppm<br>(Vacated) TWA: 9 mg/m <sup>3</sup><br>TWA: 5 ppm<br>TWA: 9 mg/m <sup>3</sup> |

| Component   | CAS No  | European Union  | The United Kingdom  | Germany                 |
|-------------|---------|-----------------|---------------------|-------------------------|
| Formic acid | 64-18-6 | TWA: 5 ppm 8 hr | STEL: 15 ppm 15 min | TWA: 5 ppm (8 Stunden). |

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|  |  |                               |                                                                                           |                                                                                                                                                                                                                                  |
|--|--|-------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | TWA: 9 mg/m <sup>3</sup> 8 hr | STEL: 28.8 mg/m <sup>3</sup> 15 min<br>TWA: 5 ppm 8 hr<br>TWA: 9.6 mg/m <sup>3</sup> 8 hr | AGW - exposure factor 2<br>TWA: 9.5 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 5 ppm (8 Stunden). MAK<br>TWA: 9.5 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 10 ppm<br>Höhepunkt: 19 mg/m <sup>3</sup> |
|--|--|-------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## ACGIH - Biological Exposure Indices

| Component   | CAS No  | ACGIH - Biological Exposure Indices |
|-------------|---------|-------------------------------------|
| Formic acid | 64-18-6 | Not listed                          |

## Exposure Controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Face protection shield or Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Chemical resistant apron. Boots. Chemical protection suit (EN 14605).

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Personal protective equipment

#### Respiratory Protection

Use only those certified by the Korea Occupational Safety and Health Administration.

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### Recommended Filter type:

Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

Appearance (Physical State, Color, etc.) Colorless Liquid

Odor

pungent

Odor Threshold

No data available

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|                              |                                |                                   |
|------------------------------|--------------------------------|-----------------------------------|
| pH                           | 2.1                            | 10 g/L aq.sol                     |
| Melting Point/Range          | 8 °C / 46.4 °F                 |                                   |
| Softening Point              | No data available              |                                   |
| Boiling Point/Range          | 101 °C / 213.8 °F              | @ 760 mmHg                        |
| Flash Point                  | 50 °C / 122 °F                 | Method - No information available |
| Evaporation Rate             | No data available              |                                   |
| Flammability (solid,gas)     | Not applicable                 | Liquid                            |
| Explosion Limits             | Lower 10 vol%<br>Upper 57 vol% |                                   |
| Vapor Pressure               | 44 mbar @ 20 °C                |                                   |
| Vapor Density                | No data available              | (Air = 1.0)                       |
| Specific Gravity / Density   | 1.220                          |                                   |
| Bulk Density                 | Not applicable                 | Liquid                            |
| Water Solubility             | Miscible                       |                                   |
| Solubility in other solvents | No information available       |                                   |

## Partition Coefficient (n-octanol/water)

| Component   | CAS No  | log Pow |
|-------------|---------|---------|
| Formic acid | 64-18-6 | -0.54   |

|                           |                          |                                        |
|---------------------------|--------------------------|----------------------------------------|
| Autoignition Temperature  | 520 °C / 968 °F          |                                        |
| Decomposition Temperature | No data available        |                                        |
| Viscosity                 | 1.47 mPa.s @ 20 °C       |                                        |
| Explosive Properties      |                          | explosive air/vapour mixtures possible |
| Oxidizing Properties      | No information available |                                        |
| Molecular Formula         | C H2 O2                  |                                        |
| Molecular Weight          | 46.02                    |                                        |

## SECTION 10: STABILITY AND REACTIVITY

Reactivity  
None known, based on information available

Chemical Stability  
Strong reducing agent. Fire and explosion risk in contact with oxidizing agents.  
Hygroscopic. heat sensitive. Decomposes to water and carbon dioxide.

Possibility of Hazardous Reactions  
Hazardous Polymerization Hazardous Reactions  
Hazardous polymerization does not occur.  
None under normal processing.

Conditions to Avoid  
Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Exposure to moist air or water.

Incompatible Materials  
Strong oxidizing agents. Metals. Finely powdered metals. Strong bases.

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## Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### Information on expected route of exposure

|            |                                                                                                                                                                                    |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation | Causes severe burns. May be harmful if inhaled. Harmful by inhalation.                                                                                                             |
| Ingestion  | Causes severe burns. May be harmful if swallowed. Ingestion causes burns of the upper digestive and respiratory tracts. Can burn mouth, throat, and stomach. Harmful if swallowed. |
| Eyes       | Causes severe burns. Causes burns. Corrosive to the eyes and may cause severe damage including blindness. Risk of serious damage to eyes.                                          |
| Skin       | Causes severe burns. May be harmful in contact with skin. Causes burns.                                                                                                            |

#### Information on Health Hazards

##### (a) acute toxicity;

|            |                   |
|------------|-------------------|
| Oral       | Category 4        |
| Dermal     | No data available |
| Inhalation | Category 3        |

| Component   | CAS No  | LD50 Oral         | LD50 Dermal       | LC50 Inhalation                |
|-------------|---------|-------------------|-------------------|--------------------------------|
| Formic acid | 64-18-6 | 730 mg/kg ( Rat ) | No data available | 7.85 mg/l (Rat) 4h<br>OECD 403 |

(b) skin corrosion/irritation; Category 1

(c) serious eye damage/irritation; Category 1

##### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

| Component   | CAS No  | Test method       | Test species      | Study result      |
|-------------|---------|-------------------|-------------------|-------------------|
| Formic acid | 64-18-6 | No data available | No data available | No data available |

(e) germ cell mutagenicity; No data available

| Component   | CAS No  | Test method       | Test species      | Study result      |
|-------------|---------|-------------------|-------------------|-------------------|
| Formic acid | 64-18-6 | No data available | No data available | No data available |

(f) carcinogenicity; No data available

| Component   | CAS No  | Test method       | Test species /<br>Duration | Study result      |
|-------------|---------|-------------------|----------------------------|-------------------|
| Formic acid | 64-18-6 | No data available | No data available          | No data available |

There are no known carcinogenic chemicals in this product

| Component   | CAS No  | IARC       | NTP        | ACGIH      | OSHA       | UK         |
|-------------|---------|------------|------------|------------|------------|------------|
| Formic acid | 64-18-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

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(g) reproductive toxicity; No data available

| Component   | CAS No  | Test method       | Test species / Duration | Study result      |
|-------------|---------|-------------------|-------------------------|-------------------|
| Formic acid | 64-18-6 | No data available | No data available       | No data available |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

## Other Adverse Effects

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

| Component   | CAS No  | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|-------------|---------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Formic acid | 64-18-6 | Applicable                               | Not applicable                                   | Not applicable                          |

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component   | CAS No  | Freshwater Fish                        | Water Flea         | Freshwater Algae   | Microtox             |
|-------------|---------|----------------------------------------|--------------------|--------------------|----------------------|
| Formic acid | 64-18-6 | Leuciscus idus: LC50 = 46-100 mg/L/96h | EC50 = 34 mg/L/48h | EC50 = 25 mg/L/96h | EC50 = 46.7 mg/L/17h |

### Persistence and degradability

#### Persistence

#### Degradation in sewage treatment plant

Readily biodegradable

Miscible with water, Persistence is unlikely, based on information available.

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### Bioaccumulative potential

Bioaccumulation is unlikely

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| Formic acid | -0.54   | 0.22 dimensionless            |

### Mobility in soil

The product is water soluble, and may spread in water systems . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

### Ozone Depletion Potential

| Component   | CAS No  | Ozone Depletion Potential |
|-------------|---------|---------------------------|
| Formic acid | 64-18-6 | Not listed                |

### Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods



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|                                            |                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).                                                                                                                                                                                              |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                   |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. |

## SECTION 14: TRANSPORT INFORMATION

### Road and Rail Transport

|                         |             |
|-------------------------|-------------|
| UN-No                   | UN1779      |
| Proper Shipping Name    | FORMIC ACID |
| Hazard Class            | 8           |
| Subsidiary Hazard Class | 3           |
| Packing Group           | II          |

### IATA

|                         |             |
|-------------------------|-------------|
| UN-No                   | UN1779      |
| Proper Shipping Name    | FORMIC ACID |
| Hazard Class            | 8           |
| Subsidiary Hazard Class | 3           |
| Packing Group           | II          |

### IMDG/IMO

|                         |                       |
|-------------------------|-----------------------|
| UN-No                   | UN1779                |
| Proper Shipping Name    | FORMIC ACID           |
| Hazard Class            | 8                     |
| Subsidiary Hazard Class | 3                     |
| Packing Group           | II                    |
| Marine Pollutant        | No hazards identified |

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**Legend:** X - Listed '-' - Not Listed

### International Inventories

| Component   | CAS No  | KECL | TSCA | EINECS    | IECSC | DSL | NDSL | PICCS | ENCS | ISHL | AICS |
|-------------|---------|------|------|-----------|-------|-----|------|-------|------|------|------|
| Formic acid | 64-18-6 | X    | X    | 200-579-1 | X     | X   | -    | X     | X    | X    | X    |

| Component   | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Formic acid | 64-18-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y34                      |

| Component | CAS No | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential |
|-----------|--------|----------|------------------------------|---------------------------|
|-----------|--------|----------|------------------------------|---------------------------|

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|-------------|---------|--------|----------------|----------------|
| Formic acid | 64-18-6 | Listed | Not applicable | Not applicable |
|-------------|---------|--------|----------------|----------------|

## Korean National Regulations

| Component   | CAS No  | Act on Registration and Evaluation of Chemical Substances (K-REACH) | Authorised Chemicals | Existing Substances Subject to Registration |
|-------------|---------|---------------------------------------------------------------------|----------------------|---------------------------------------------|
| Formic acid | 64-18-6 | Annex 1 - KE-17233                                                  | Not applicable       | Listed                                      |

| Component   | CAS No  | Chemical Control Act - Toxic Chemicals | Chemical Control Act - Prohibited Chemicals | Chemical Control Act - Use Restricted Chemicals |
|-------------|---------|----------------------------------------|---------------------------------------------|-------------------------------------------------|
| Formic acid | 64-18-6 | 2022-1-1091 (>10%)                     | Not applicable                              | Not applicable                                  |

| Component   | CAS No  | Chemical Control Act - Accident Precaution Chemicals (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures) |
|-------------|---------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Formic acid | 64-18-6 | >25%                                                                 | 20000 kg/yr                                                                                    | 1500000 kg/yr                                                                                          |

| Component   | CAS No  | Waste Control Law | Ministry of Environment - CMR risk | Ministry of Environment - Critically Controlled Substance |
|-------------|---------|-------------------|------------------------------------|-----------------------------------------------------------|
| Formic acid | 64-18-6 | > 10% (CCA)       | Not applicable                     | Not applicable                                            |

| Component   | CAS No  | ISHA - Harmful Agents Subject to Work Environment Monitoring | ISHA - Prohibited substances | ISHA - Substances requiring permission |
|-------------|---------|--------------------------------------------------------------|------------------------------|----------------------------------------|
| Formic acid | 64-18-6 | Listed                                                       | Not applicable               | Not applicable                         |

| Component   | CAS No  | ISHA - Substances subject to control | ISHA - Harmful Agents Requiring Health Examination | ISHA - Permissible Exposure Limits |
|-------------|---------|--------------------------------------|----------------------------------------------------|------------------------------------|
| Formic acid | 64-18-6 | Listed                               | Not applicable                                     | Not applicable                     |

| Component   | CAS No  | ISHA - Subject to Process Safety Reports (minimum quantity) | ISHA - Threshold Limit Values (TLVs) Chemicals | ISHA - Special management materials |
|-------------|---------|-------------------------------------------------------------|------------------------------------------------|-------------------------------------|
| Formic acid | 64-18-6 | 5000 kg                                                     | TWA: 5 ppm                                     | Not applicable                      |

## National Fire Association - Dangerous Substances Minimum quantity requiring a permit

| Component   | CAS No  | Class 1 - Oxidising solids | Class 2 - Flammable solid | Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet | Class 4 - Flammable liquids           | Class 5 - Self-reactive substances | Class 6 - Oxidising liquids |
|-------------|---------|----------------------------|---------------------------|----------------------------------------------------------------------------------|---------------------------------------|------------------------------------|-----------------------------|
| Formic acid | 64-18-6 | Not applicable             | Not applicable            | Not applicable                                                                   | 4. Group 2 Petroleum (Soluble) 2000 L | Not applicable                     | Not applicable              |

## Control Parameters

| Component   | CAS No  | Korea                                  | ACGIH - Biological Exposure Indices |
|-------------|---------|----------------------------------------|-------------------------------------|
| Formic acid | 64-18-6 | TWA: 5 ppm<br>TWA: 9 mg/m <sup>3</sup> | Not listed                          |

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## US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

| Component   | CAS No  | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|-------------|---------|----------------------------------|----------------------------|
| Formic acid | 64-18-6 | Not applicable                   | Not applicable             |

## CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355)

| Component   | CAS No  | CERCLA Extremely Hazardous Substances RQs | Hazardous Substances RQs | SARA 313 - Threshold Values % |
|-------------|---------|-------------------------------------------|--------------------------|-------------------------------|
| Formic acid | 64-18-6 | Not applicable                            | 5000 lb                  | 1.0                           |

## GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Danger.

H226 - Flammable liquid and vapor. H302 - Harmful if swallowed. H314 - Causes severe skin burns and eye damage. H331 - Toxic if inhaled. EUH071 - Corrosive to the respiratory tract.

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P280 - Wear protective gloves/protective clothing/eye protection/face protection. P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P310 - Immediately call a POISON CENTER or doctor/physician.

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

### Training Advice

# SAFETY DATA SHEET

Formic acid, OPTIMA LC/MS Grade (99.5%)

Revision Date 06-Jun-2024

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

|                  |                       |
|------------------|-----------------------|
| Creation Date    | 02-Nov-2009           |
| Revision Date    | 06-Jun-2024           |
| Revision Number  | 6                     |
| Revision Summary | SDS sections updated. |

## MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**